Battery Specifications

Executive Summary

The following specifications outline a custom-designed LiFePO4 battery pack intended to serve as the primary power source for a Brushless DC Motor (BLDCM) system.

Nominal Voltage	$51.2V$
Total Capacity	$42Ah\ (\approx 2150Wh)$
System Max Continuous Discharge	$60A(LimitedbyBMS)$
Cell-Level Max Continuous Discharge	$126A$
Cell Chemistry	Lithium Iron Phosphate $(LiFePO_4)$
Expected Life Cycle	$> 2000Cycle$
Series/Parallel Configuration	16S7P
Cooling Method	Passive

System Requirements

The battery pack is designed to power a BLDCM with the following key specifications:

Nominal Voltage	48V
Rated Speed	3000 RPM
Rated Power	660W

The system requires two hours of continuous operation at full load. Assuming a combined electromechanical conversion efficiency (inverter and motor) of 80%, the required energy capacity is calculated as follows:

$$E_{req} = \frac{660W}{80}$$

Consequently, the required capacity in Ampere-hours (Ah) at the nominal system voltage is:

$$Ah_{req} = \frac{1650\ Wh}{48V} \approx 34.38\ Ah$$

Battery Design

Chemistry Selection

Lithium Iron Phosphate (LiFePO4) was selected based on the following engineering parameters:

1. **Thermal Stability:** Inherently stable chemistry minimizes the risk of thermal runaway events.
2. **Cycle Life:** Capable of $> 2000\ charge/discharge\ cycles$, providing a longer operational lifespan compared to alternative lithium chemistries.
3. **Environmental Impact:** Free of toxic heavy metals.
4. **Environmental Tolerance:** Maintains reliable performance across a wide operating temperature range.
5. **Total Cost of Ownership:** Extended lifespan and reduced maintenance requirements yield long-term cost efficiency.

Cell Specifications

Form Factor	32700
Nominal Voltage	3.2V
Nominal Capacity	6 Ah
Max Continuous Discharge Current	18 A

Electrical Design

To size the battery pack appropriately, the required capacity must account for the safe Depth of Discharge (DoD) limit of the cells. Assuming a strict DoD limit of 90% (0.9) to preserve cell longevity:

$$Ah_{pack} = \frac{Ah_{req}}{DoD_{limit}}$$

$$Ah_{pack} = \frac{34.38\ Ah}{0.9} = 38.2\ Ah$$

Parallel Configuration (N_p):

Given each cell's capacity of 6 Ah, the required number of parallel cells is:

$$N_p = \frac{38.2}{6} = 6.37$$

Rounding up to ensure adequate capacity, N_p is designated as 7.

Series Configuration ($\backslash mathbitN_{\backslash mathbit}s$):

The motor's nominal voltage requirement is 48 V. Based on the 3.2 V nominal voltage of each cell:

$$N_{s\,(base)} = \frac{48}{3.2} = 15$$

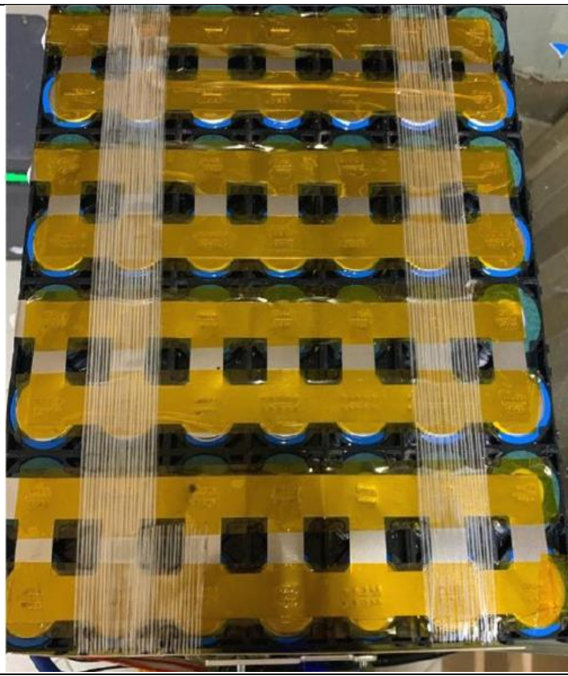
While 15S yields exactly 48V, a 16S configuration is standard for 48V-class systems. Increasing N_s to 16 accounts for voltage sag under full load, compensates for resistive losses across system wiring and inverter switching.

Final Pack Specifications (16S7P):

Nominal Voltage	$16 \times 3.2\ V = 51.2\ V$
Nominal Capacity	$7 \times 6\ Ah = 42\ Ah\ (yielding \approx 2150\ Wh)$
Max Continuous Discharge Current	$7 \times 18\ A = 126\ A$

Implementation Details

The final physical assembly utilizes the designed 16S7P configuration using EPRO 32700 LiFePO4 cells.



Battery Management System (BMS):

The pack is governed by a JIABAIDA DB24SA03 BMS.

- **Hardware Capabilities:** Supports up to 24 series strings.
- **Current Limiting:** Features a maximum continuous current rating of 60 A. **Note:** This acts as the primary system bottleneck; the pack cannot output the cell-capable 126 A without triggering over-current protection. However, 60 A is highly sufficient for the continuous $\approx 18\ A$ draw of the target load.
- **Telemetry & Communication:** Features Bluetooth connectivity for mobile application monitoring and a UART port for external communication protocols.


UART
B+
H+ H− B+
C−
JIABAIDA
DB24SA03 V1.0 24S 60A
Lifepo4 3.2v UART
Common port with balance
PN: 18300005 SN248244267
Step1B−
24
BCO
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